

Structured Loss Amplification for U-Net-based Human-scale Dark-field CT Streak Reduction



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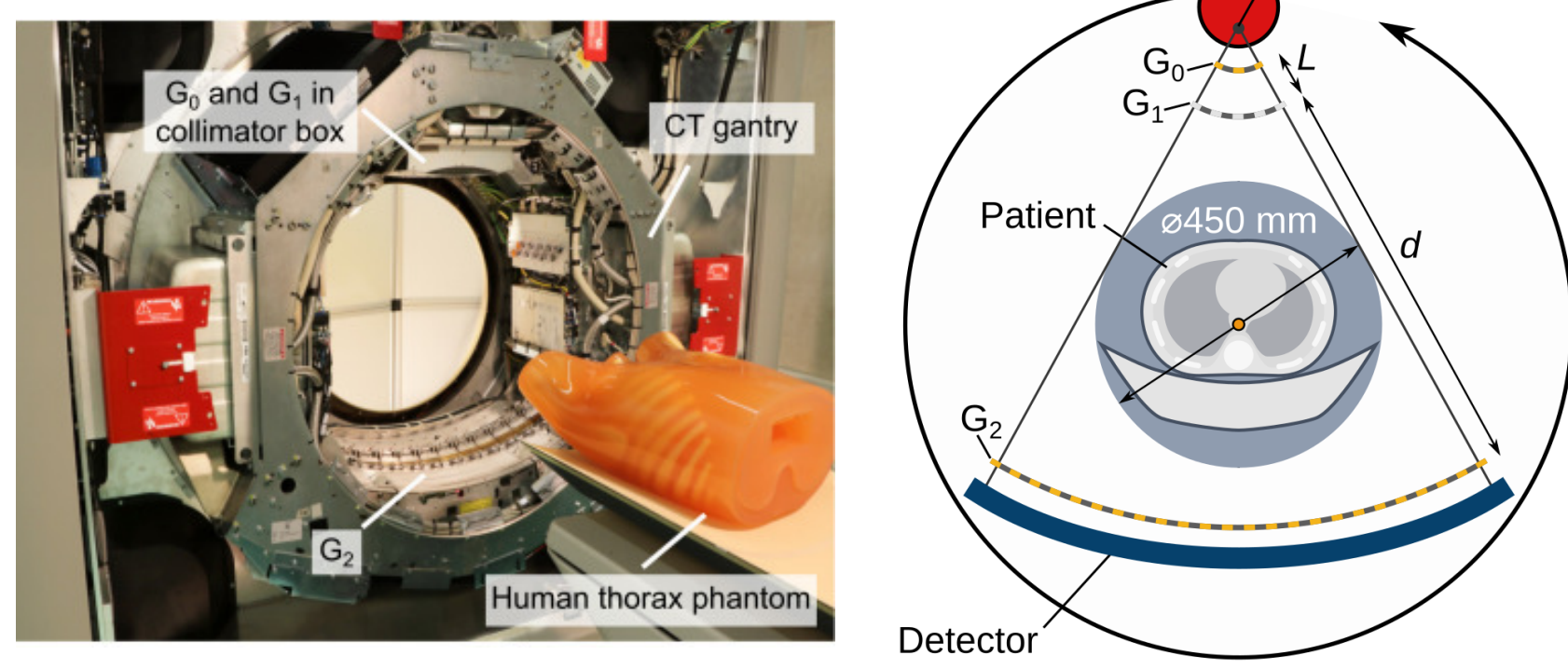
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INTRODUCTION

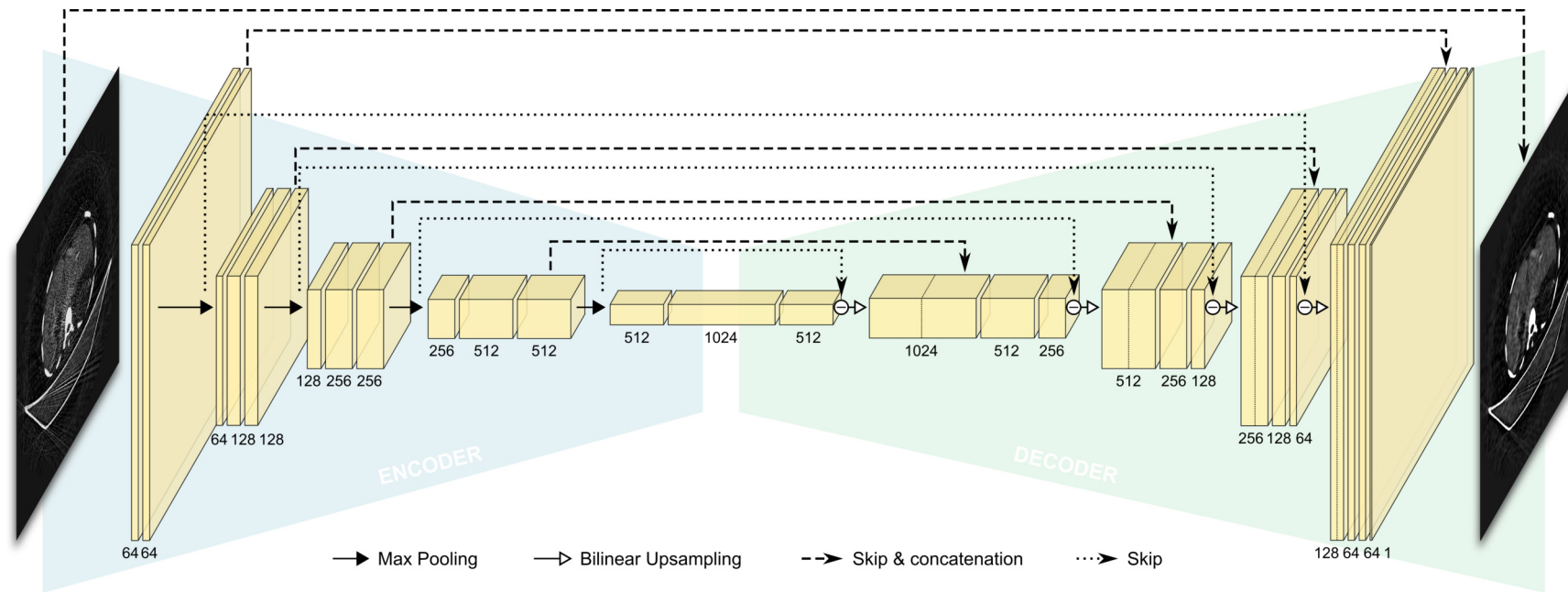
- **Dark-field Computed Tomography (DFCT)** detects lung alterations via small-angle X-ray scattering [1]
- Human-scale DFCT [2] image quality is limited due to **streak artifacts** from insufficient phase retrieval [3, 4]
- **Convolutional neural networks (CNNs)** enable image enhancement using multi-rotation ground-truth (GT) data as target [5]

METHODS

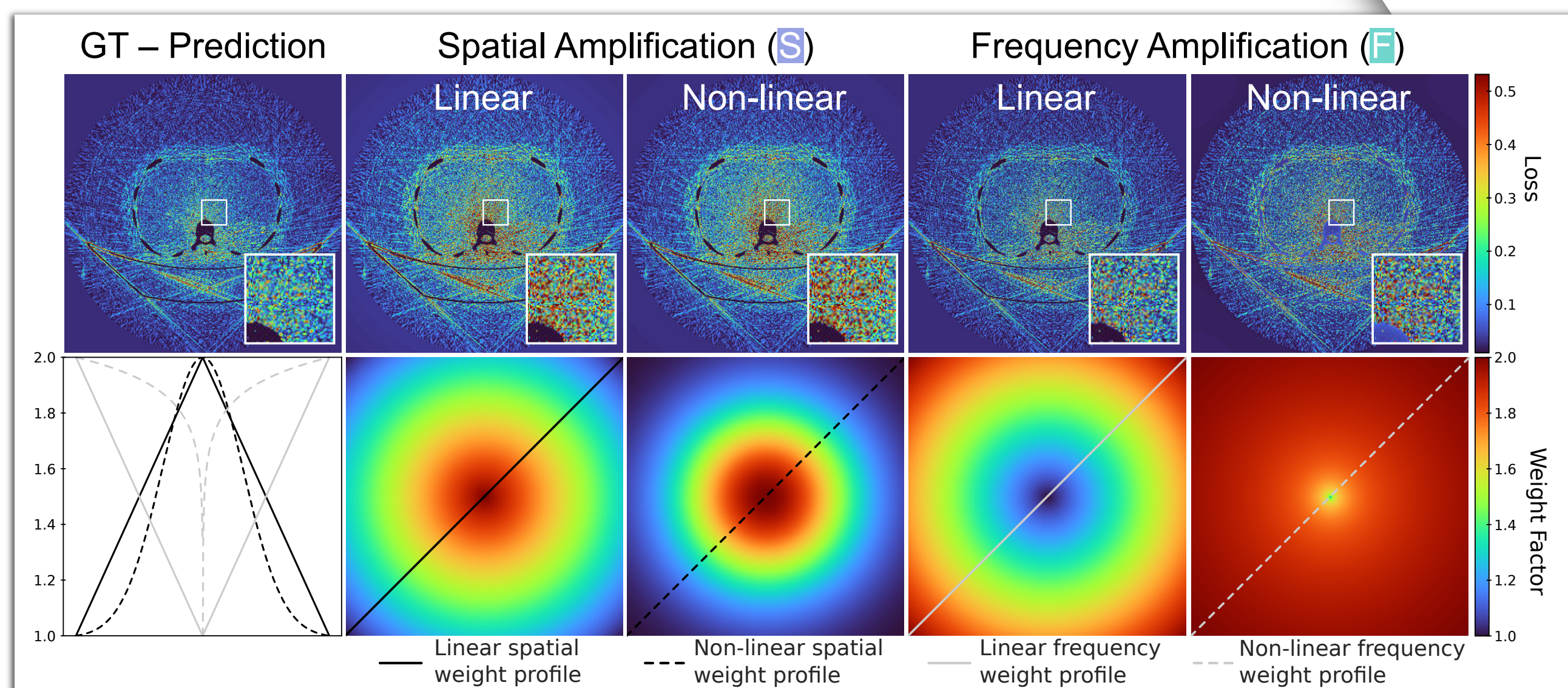
- Prototype gantry with rotating **Talbot-Lau interferometer**



- **2D U-Net** [6] trained on single-rotation scans (input) at clinically feasible dose and multi-rotation scans (GT) at up to 20-fold dose

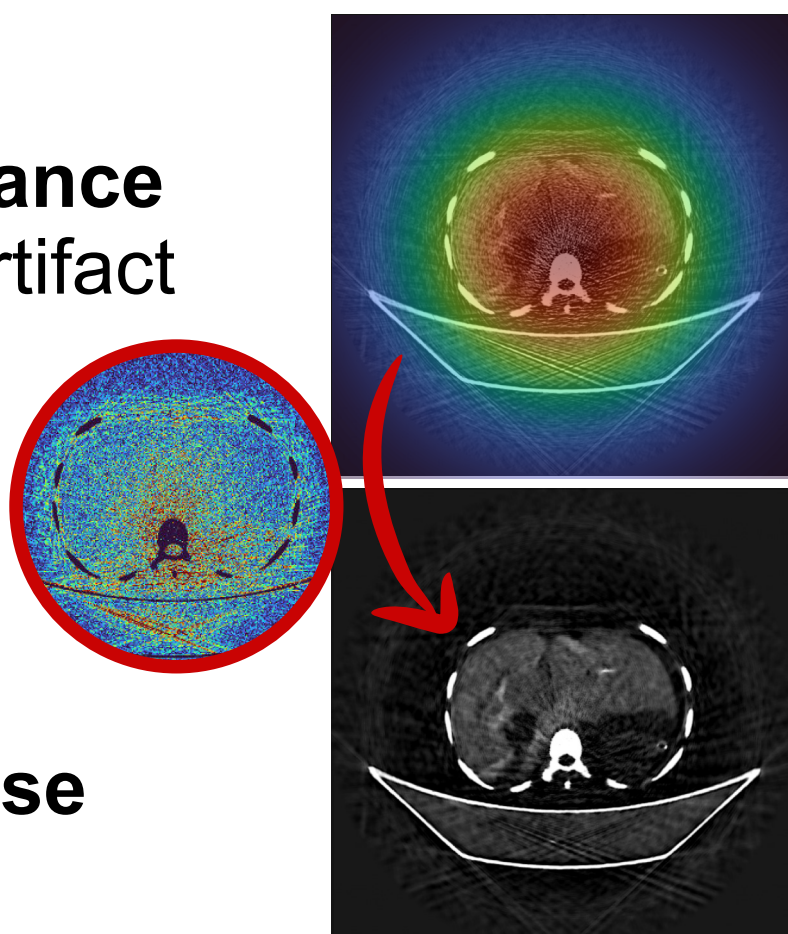


- **Structured loss amplification** boosts loss map with designed spatial (S) and frequency (F) weights



CONCLUSION & OUTLOOK

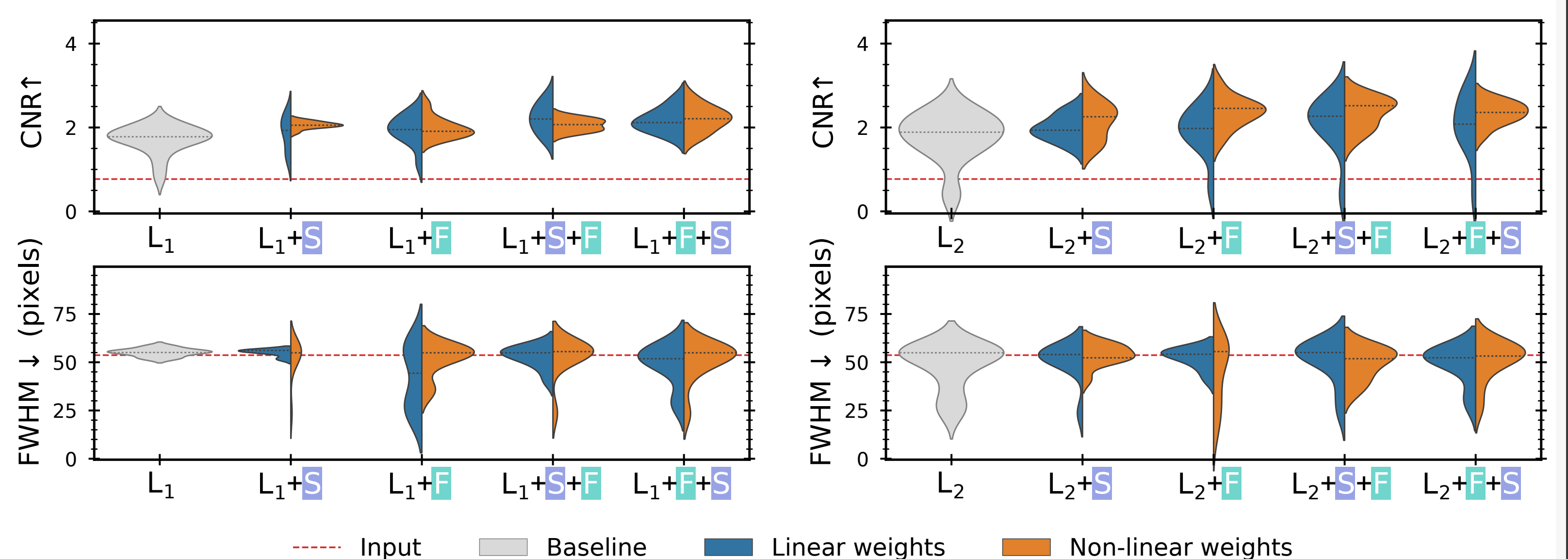
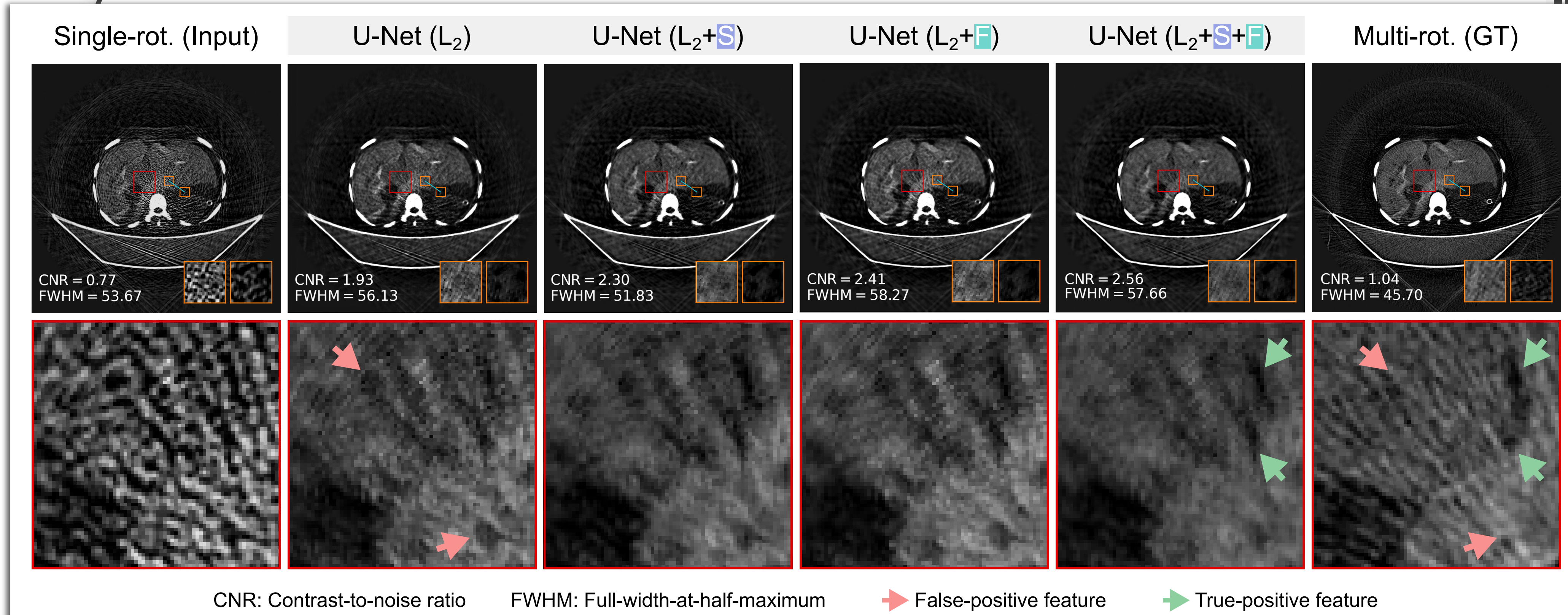
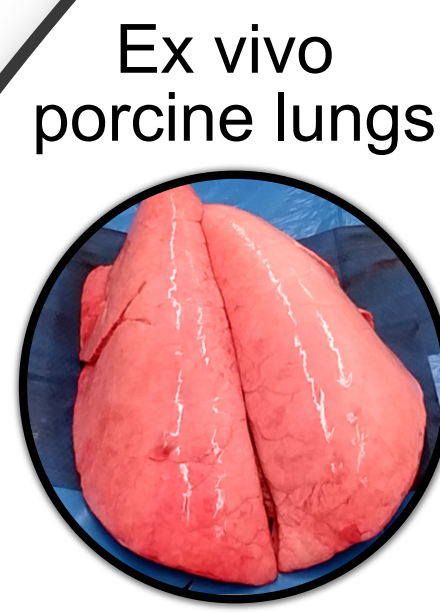
- Structured loss amplification **improves performance and robustness** of U-Net-based DFCT streak artifact reduction
- Formulation allows easy integration of **prior knowledge** into training objective
- U-Net suppresses both **streak artifacts and noise**



- Weighting schemes taking non-local, directed **signature of streaks** into account
- **GT enhancement** strategies to remove residual streaks in multi-rotation scans
- **Reader study** to validate improved DFCT quality for pulmonary imaging

RESULTS

- **Spatial amplification (S)** with central weights increases both CNR and FWHM
- **Frequency amplification (F)** with high-frequency weights enhances CNR at slightly worse FWHM
- **Combined spatio-frequency amplification (S+F)** provides highest CNR at similar FWHM compared to baseline
- **CNR up to 33% higher** (2.52 ± 0.37 , $n = 10$) compared to baseline (1.89 ± 0.58) using L_2 loss



- Model evaluation across $n = 10$ training runs for each configuration demonstrates robustness of observed performance gains

REFERENCES

- [1] Pfeiffer et al. (2008), *Nature Materials*
- [2] Viermetz et al. (2022), *PNAS*
- [3] Schmid et al. (2023), *IEEE TMI*
- [4] Haeusele et al. (2024), *IEEE TMI*
- [5] Kumschier et al. (2024), *Med Phys*
- [6] Ronneberger et al. (2015), *MICCAI*

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